LAB 7: Fuzzing for Input Validation Bugs

Name : Sohel Mahibubasa Chapparband

SRN : PES2UG25CS826

SEC : F

SEM : 5

Installing Required Dependencies

installation Commands:

python -m pip insItall hypothesis

python -m pip install pytest

Running the Original Buggy Application

```
PS C:\Users\HP\Desktop\SE\Lab 7\L6_L7_62_attendance_eligibility-main\LAB7> python attendance_eligibility_parser.py
==== Running normalize_srn ====
Enter an SRN (e.g. pes1ug23cs001): PES2UG25CS826
Normalized SRN: PES2UG25CS826

==== Running attendance_from_ratio ====
Enter classes attended and held as ATTENDED/HELD (e.g. 30/40): 45/60
Attendance percent: 75

==== Running attendance_from_register ====
Enter the register as P/A marks (e.g. PPAPP): PPAAP
Attendance percent: 60
```

Completing Hypothesis Test Cases

```
PS C:\Users\HP\Desktop\SE\Lab 7\L6_L7_62_attendance_eligibility-main\LAB7> python -m pytest test_attendance_eligibility_pa
rser.py
    | Failing test case: test_attendance_from_register_no_crash(
    |     s=None,
    | )
    +------------------------------------
========================================== short test summary info ==========================================
FAILED test_attendance_eligibility_parser.py::test_normalize_srn_no_crash - AssertionError: normalize_srn crashed with: No
ne
FAILED test_attendance_eligibility_parser.py::test_attendance_from_ratio_no_crash - ExceptionGroup: Hypothesis found 4 dis
tinct failures. (4 sub-exceptions)
FAILED test_attendance_eligibility_parser.py::test_attendance_from_register_no_crash - ExceptionGroup: Hypothesis found 2
distinct failures. (2 sub-exceptions)
========================================== 3 failed in 2.83s ==========================================
```

Running the Fixed Application

```
PS C:\Users\HP\Desktop\SE\Lab 7\L6_L7_62_attendance_eligibility-main\LAB7> python attendance_eligibility_parser.py
==== Running normalize_srn ====
Enter an SRN (e.g. pes1ug23cs001): PES2UG25CS826
Normalized SRN: PES2UG25CS826

==== Running attendance_from_ratio ====
Enter classes attended and held as ATTENDED/HELD (e.g. 30/40): 45/60
Attendance percent: 75

==== Running attendance_from_register ====
Enter the register as P/A marks (e.g. PPAPP): PPAAP
Attendance percent: 60
```

Testing the Fixed Application

```
PS C:\Users\HP\Desktop\SE\Lab 7\L6_L7_62_attendance_eligibility-main\LAB7> python -m pytest test_attendance_eligibility_parser.py
============================================= test session starts =============================================
platform win32 -- Python 3.12.10, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\HP\Desktop\SE\Lab 7\L6_L7_62_attendance_eligibility-main\LAB7
plugins: hypothesis-6.170.15
collected 3 items

test_attendance_eligibility_parser.py ...                                                        [100%]

============================================== 3 passed in 1.37s ==============================================
```